

XIXIANG (SAM) LUO

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EDUCATION

University of Michigan - Ann Arbor

B.S. Computer Science

Ann Arbor, MI

GPA: 3.7/4.0 | Expected May 2027

EXPERIENCE

WEX

Remote

Enterprise Applications Engineer Intern

May 2026 – Aug 2026

- Built a self-service onboarding workflow for OWI, WEX's centralized identity and authentication platform, automating Auth0, MongoDB, and HashiCorp Vault provisioning and reducing new application integration time from 1 week to 5 minutes.
- Improved WEX's customer service AI assistant's card management skill across 3 user intents with 9 agent tools, increasing task success rate by 15% through stronger prompt and code level guardrails.
- Integrated WEX's eGain knowledge base for real-time, mid-workflow Q&A without losing progress during multi-turn intents.
- Hardened agent workflows with 14 DeepEval scenarios and 25 unit tests for end-to-end validation, using structured logging and Arize Phoenix tracing to rapidly diagnose and resolve tool-call and workflow failures.
- Built a GitHub Actions CI/CD pipeline for regular releases, automating lower-environment deployments with AWS AgentCore and defining container image promotion logic from staging to production.

Tera - SHUTTL LLC

Remote

Software Developer Intern

Dec 2024 – Apr 2025

- Developed the user profile page and an administrative dashboard with 15+ reusable Svelte components, integrated 17+ RESTful API endpoints, and implemented secure file uploads using AWS S3 storage buckets.
- Optimized user pagination queries by analyzing PostgreSQL slow query logs and applying a subquery strategy with a covering index, reducing latency by 30%.
- Implemented SMS-based two-factor authentication with Twilio, securing protected routes through one-time code validation and building the user interface with Shadcn components.

PROJECTS

Distributed KV Storage with Sharding

Go, RPC, Paxos, Replication, Sharding

- Built a strongly consistent, fault-tolerant, in-memory key-value store in Go that handled concurrent read/write requests using RPC and Paxos-based replication.
- Increased throughput and scalability through sharding, distributing the key space across multiple Paxos replica groups so independent groups could process requests in parallel.
- Designed an algorithm to rebalance shards evenly across groups with minimal shard movement during joins, leaves, and shard reassignment, while maintaining linearizability under concurrent writes.
- Implemented at-most-once semantics through duplicate-request detection, ensuring retried client operations were not executed multiple times during failures or reconfiguration.

Transformer Language Model From Scratch

Python, PyTorch, BPE, Transformers, AdamW

- Built an end-to-end Transformer language modeling pipeline from scratch, including tokenization, model architecture, loss, optimizer, training, checkpointing, and evaluation.
- Implemented a Byte-Pair Encoding tokenizer to train custom vocabularies and convert text datasets into token ID sequences.
- Developed core Transformer components in PyTorch, including embeddings, attention, feedforward layers, normalization, cross-entropy loss, and AdamW optimization.
- Trained and evaluated autoregressive language models on large-scale text corpora, generating samples and measuring model quality with validation loss and perplexity.

Tlias | Spring Boot, Vue 3, MyBatis, JWT, Docker

github.com/luoxixiang23921/vue-tlias-management

- Built an educational management system with Spring Boot and Vue 3 using a three-tier architecture.
- Implemented paginated search with MyBatis dynamic queries for student and employee records.
- Added JWT authentication and AOP-based operation logging to improve security and auditability.

SKILLS & TECHNOLOGIES

Languages: Java, Python, Go, TypeScript, JavaScript, C++, C, SQL, Bash, HTML/CSS

ML & AI: PyTorch, Transformers, Hugging Face, Scikit-Learn, LangChain, LangGraph, RAG, NLP

Web & Backend: REST API, Spring Boot, ReactJS, Next.js, FastAPI, Node.js, PostgreSQL, MySQL, MongoDB, Kafka, Redis

Tools & Cloud: Git, AWS (Lambda, EC2, S3, DynamoDB, RDS), Docker, Kubernetes, CI/CD, Grafana, Splunk, OpenTelemetry, Terraform, Nginx